

MORNING BRIEF AGENT

Technical Build Plan — Free Sources Edition

Azure · Azure AI Foundry · Claude claude-opus-4-20250514 · Twilio · 100% Free Data Sources

WHAT CHANGED IN THIS VERSION

All paid and paywalled sources have been removed (WSJ, FT, The Information, Stratechery, Morning Brew). They have been replaced with 26 fully free sources spanning RSS feeds, public APIs (Reddit, Hacker News, Bluesky, YouTube), GitHub Trending, and ProductHunt. Twitter/X API has also been replaced with Bluesky public API — same crowd, zero cost. Total monthly infrastructure cost drops to ~\$8/month.

1. Project Overview

An autonomous morning intelligence agent that collects content from 26 free sources every day at 6:00 AM PST, runs it through Claude claude-opus-4-20250514 on Azure AI Foundry, and delivers a structured long-form digest via SMS by 6:30 AM PST. Zero paid data subscriptions required.

Parameter	Value
Scrape Time 6:00 AM PST daily	Delivery Time 6:30 AM PST daily
Cloud Platform Microsoft Azure	LLM Claude claude-opus-4-20250514 via Azure AI Foundry
Delivery Channel SMS via Twilio	Fallback WhatsApp Business API (Twilio)
Data Source Cost \$0/month — all free	Total Source Count 26 sources (12 econ + 14 tech)
Digest Format Long-form structured digest	Sections Economy + Tech + Watch Today + Founder Angle

2. System Architecture

Same 5-stage pipeline as the original plan. The only change is Stage 2 — the ingestion layer now uses exclusively free endpoints.

Stage	Component	Azure Service / Tool
1 — Trigger	Cron Scheduler (6:00 AM PST)	Azure Container Apps Jobs
2 — Ingest	RSS + Public API Fetcher	Python: feedparser, httpx, BeautifulSoup — free endpoints only
3 — Process	LLM Summarizer & Editor	Azure AI Foundry → Claude claude-opus-4-20250514
4 — Format	Digest Formatter	Python template engine (same container)
5 — Deliver	SMS Sender	Twilio SDK → your phone number

3. Free Data Sources (26 Total)

All sources below are 100% free to access programmatically. Green = RSS feed (simplest). Blue = free public API. Orange = lightweight public scrape (no auth, no ToS violation for personal use).



Economy & Business — 12 Free Sources

RSS feeds + Reddit public API + NPR + AP Wire

Source	Method	Endpoint / Notes
Bloomberg	RSS — Free	feeds.bloomberg.com/markets/news.rss · Free headlines (no full text)
Reuters	RSS — Free	feeds.reuters.com/reuters/businessNews · Fully free, excellent global coverage
CNBC	RSS — Free	cnbc.com/id/100003114/device/rss/rss.html · Markets + business
MarketWatch	RSS — Free	marketwatch.com/rss/topstories · Real-time market moves
Yahoo Finance	RSS — Free	finance.yahoo.com/news/rssindex · High volume, good signal-to-noise after filtering
AP Business News	RSS — Free	feeds.apnews.com/rss/business · Wire-quality, no paywall ever
Federal Reserve	RSS — Free	federalreserve.gov/feeds/press_all.xml · Official Fed press releases

The Economist (free)	RSS — Free	economist.com/finance-and-economics/rss.xml · Free articles only (~2-3/week)
r/economics	Free Reddit API	reddit.com/r/economics/top.json?limit=5&t=day · Top posts daily
r/investing	Free Reddit API	reddit.com/r/investing/top.json?limit=5&t=day · Retail investor pulse
Planet Money (NPR)	RSS — Free	feeds.npr.org/510289/podcast.xml · Free always
Axios Markets	RSS — Free	axios.com/feeds/feed.rss (filter economy tag) · Free tier covers headlines



Tech & AI — 14 Free Sources

RSS + Hacker News API + Reddit + GitHub + Bluesky + YouTube RSS

Source	Method	Endpoint / Notes
The Verge	RSS — Free	theverge.com/rss/index.xml · Always free
Ars Technica	RSS — Free	feeds.arstechnica.com/arstechnica/index · Always free, deep technical
TechCrunch	RSS — Free	techcrunch.com/feed · Free, strong on funding/startups
WIRED	RSS — Free	wired.com/feed/rss · ~10 free articles/month via RSS
Hacker News	Free Public API	hacker-news.firebaseio.com/v0/topstories.json · Top 30 stories daily
MIT Technology Review	RSS — Free	feeds.feedburner.com/mit-technology-review · Free tier headlines
a16z Blog	RSS — Free	a16z.com/feed · Always free, VC thesis signal
Platformer	RSS — Free	platformer.news/feed · Free tier available
r/MachineLearning	Free Reddit API	reddit.com/r/MachineLearning/top.json?limit=5&t=day · Research paper drops
r/technology	Free Reddit API	reddit.com/r/technology/top.json?limit=5&t=day · Broad tech pulse
GitHub Trending	Free Scrape	github.com/trending — scrape daily trending repos · Zero auth needed
ProductHunt	RSS — Free	producthunt.com/feed · Daily top products launched
Bluesky (Tech feed)	Free Public API	public.api.bsky.app — @karpathy.bsky.social etc. · No auth for public posts
YouTube RSS (channels)	RSS — Free	youtube.com/feeds/videos.xml?channel_id=CHANNEL_ID · @karpathy, @Fireship, @lexfridman



SOCIAL MEDIA STRATEGY — BLUESKY OVER X

The Twitter/X API costs \$100/month minimum for Basic tier — not worth it. Bluesky's public API is completely free and unauthenticated for reading public posts. The tech/AI crowd (ex-Twitter researchers, founders, journalists) has largely migrated there. Key accounts to follow: @karpathy.bsky.social, @emollick.bsky.social, @benedictevans.bsky.social. For econ/finance voices, Reddit r/economics and r/investing surfaces the same signal for free.

4. LLM Processing via Azure AI Foundry

4.1 Model Setup

Deploy Claude claude-opus-4-20250514 via Azure AI Foundry (available in the Azure Marketplace model catalog). Store the endpoint URL and API key in Azure Key Vault. The Python container calls this endpoint using the standard Anthropic SDK with the `azure_endpoint` parameter.

4.2 Prompt Strategy — Digest Structure

The system prompt instructs Claude to act as a senior analyst and produce a clean, opinionated daily brief in this exact structure:

SYSTEM PROMPT TEMPLATE

You are a senior analyst and editorial director producing a daily intelligence digest for a tech entrepreneur and investor. Be direct, opinionated, and specific. Never pad. Signal over noise.

Structure (output exactly this format):

 **MORNING BRIEF — {DATE}**

————  **ECONOMY & MARKETS** ————

3-5 bullets. Lead with the most market-moving item. Bold topic label + 1-2 sentence summary + entrepreneur implication.

————  **TECH & AI** ————

3-5 bullets. Prioritize: model releases, funding rounds >0M, architecture shifts, chip/infra, new dev tools going viral on HN/GitHub.

————  **WHAT TO WATCH TODAY** ————

2-3 items: scheduled events, earnings, regulatory decisions, developing stories breaking today.

————  **FOUNDER ANGLE** ————

1 paragraph. Synthesize today's news into one actionable insight or question for a tech entrepreneur. Be bold and opinionated.

4.3 Token Management

- Cap each source at 600 words — prioritize newest items, trim older ones
- Target input context: ~10,000 tokens (26 sources × ~400 tokens average after trimming)
- Set max_tokens = 1,500 on the completion — sufficient for a rich 4-section digest
- Estimated cost per run: ~\$0.10–0.20 — about \$3–6/month at daily cadence

5. Message Delivery via Twilio

5.1 SMS — Recommended Setup

Twilio SMS is the simplest delivery mechanism. Purchase a US number (~\$1/mo), store credentials in Azure Key Vault, and call `client.messages.create()` with the formatted digest. Twilio auto-splits messages longer than 1,600 characters into multiple SMS segments.

1. Create Twilio account at [twilio.com](https://www.twilio.com) — purchase a US long code number (~\$1/month)
2. Store `TWILIO_ACCOUNT_SID`, `TWILIO_AUTH_TOKEN`, and `FROM_NUMBER` in Azure Key Vault
3. Python delivery module calls the Twilio SDK with the formatted digest text
4. Cost: ~\$0.01/day for a 3-5 segment SMS — negligible



FORMATTING NOTE

SMS does not support markdown. The LLM output must use plain Unicode characters for visual structure: `———` dividers, `•` bullets, emoji. The prompt template above already enforces this. If you later switch to WhatsApp, the developer adds a `FORMAT_MODE` flag and the prompt switches to `*bold*` and `_italic_` WhatsApp markdown syntax.

6. Azure Infrastructure

Service	Purpose	Notes
Azure Container Apps Jobs	Run agent on schedule	Cron: 0 14 * * * (6AM PST). Pay-per-execution — ~\$0/mo at this scale
Azure Container Registry	Store Docker image	Private registry for the Python container
Azure AI Foundry	Host Claude claude-opus-4-20250514	Managed endpoint with Azure auth — billed per token
Azure Key Vault	Store all secrets	Twilio creds, Bluesky handle, Reddit client ID
Azure Log Analytics	Monitoring & alerting	Alert if job fails — email or Teams notification
Azure Blob Storage	Optional: cache digests	Store previous digest for deduplication logic

7. Code Structure

```
morning-brief-agent/
├── Dockerfile
├── requirements.txt          # feedparser, httpx, beautifulsoup4, anthropic,
twilio                       # entrypoint - orchestrates all stages
├── main.py                  # loads secrets from Azure Key Vault
├── config.py
├── ingestion/
│   ├── rss_fetcher.py      # RSS feeds (Bloomberg, Reuters, Verge, etc.)
│   ├── reddit_fetcher.py   # Reddit free JSON API (no auth needed)
│   ├── hn_fetcher.py       # Hacker News Firebase API
│   ├── bluesky_fetcher.py  # Bluesky public API - no auth for public posts
│   ├── github_trending.py  # Scrape github.com/trending (simple HTML parse)
│   └── youtube_rss.py      # YouTube channel RSS feeds
├── processing/
│   └── llm_client.py        # Azure AI Foundry → Claude
├── claude-opus-4-20250514 call
│   └── prompt_builder.py   # Assembles all scraped content into the prompt
├── delivery/
│   └── twilio_sender.py    # Sends SMS (or WhatsApp with FORMAT_MODE flag)
├── utils/
│   └── logger.py           # Structured logging → Azure Log Analytics
```

8. Phased Build Plan

Three phases, each producing a working testable deliverable.

PHASE 1 — Foundation & Happy Path	Week 1
- Set up Azure resource group, Container Registry, Key Vault - Deploy Claude claude-opus-4-20250514 on Azure AI Foundry — verify endpoint is callable - Build <code>rss_fetcher.py</code> — pull from 5 RSS sources (Reuters, CNBC, The Verge, TechCrunch, Ars Technica) - Wire into <code>llm_client.py</code> — send content to Claude, receive digest - Build <code>twilio_sender.py</code> — send a test SMS to your number - End-to-end test: scrape → summarize → receive SMS on your phone	
PHASE 2 — All 26 Sources + Prompt Tuning	Week 2

11. Add remaining RSS sources (12 econ + remaining tech)
12. Build reddit_fetcher.py — Reddit public JSON API (no credentials needed)
13. Build hn_fetcher.py — Hacker News Firebase API top stories
14. Build bluesky_fetcher.py — public API for key tech accounts (Karpathy, etc.)
15. Add github_trending.py and youtube_rss.py
16. Tune prompt daily — run for 5 days, adjust section balance, tone, and length
17. Add deduplication — skip stories already in yesterday's digest
18. Build Dockerfile, push to Azure Container Registry

PHASE 3 — Production Hardening

Week 3

19. Configure Azure Container Apps Job cron (0 14 * * * UTC = 6AM PST)
20. DST handling — update to 0 13 * * * March–November for PDT
21. Set up Azure Log Analytics — capture every run with success/failure + source counts
22. Add error alerting — notify if job fails (no SMS that morning)
23. Add retry logic — failed source = log and skip, do not crash full run
24. Run for 7 consecutive days — sign off on production readiness

9. Monthly Cost — Free Sources Edition

Item	Est. Monthly	Notes
26 Data Sources	\$0	100% free — RSS, public APIs, public scrapes
Azure Container Apps Job	~\$0	Pay-per-execution; 30 runs/month is essentially free
Azure AI Foundry (Claude claude-opus-4-20250514)	~\$3–6	~\$0.10–0.20/run × 30 days
Azure Key Vault + Log Analytics	~\$2	Minimal usage tier
Twilio SMS	~\$1–2	\$1/mo phone number + ~\$0.01/day SMS segments
TOTAL	~\$5–8/month	No paid data subscriptions — ever

10. Open Questions for Developer Kickoff

Resolve these before or during Phase 1:

25. What is your phone number for SMS delivery? Store in Key Vault — never in code.
26. Do you have an existing Azure subscription, or does one need to be created?
27. Which Bluesky accounts do you want monitored? The developer needs a list of handles (e.g. @karpathy.bsky.social).
28. Do you want GitHub Trending filtered to specific languages (Python, JavaScript, etc.) or all languages?
29. Should the digest also be stored anywhere beyond SMS — Notion, email, Slack?
30. Do you want a simple web URL where you can view today's digest in a browser as well as receiving it by SMS?

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